Sensorimotor Rhythm or Aggregate Power? An Interpretability Audit of EEGPT on Motor Imagery

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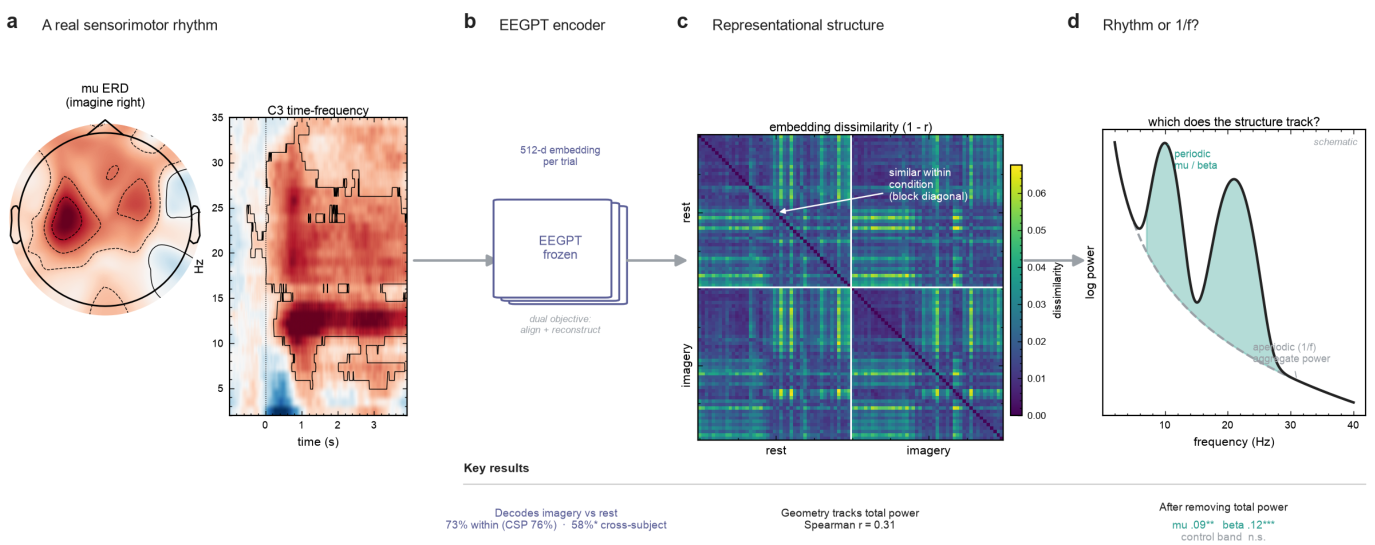
EEGPT decodes motor imagery as well as classical baselines, yet its representation is organized primarily by aggregate signal power rather than the sensorimotor rhythm, extending an aperiodic bias reported for reconstruction based foundation models to an architecture that adds representation alignment.

Introduction. A rapid wave of 2026 audits has converged on one concern about EEG foundation models: reconstruction based models encode the aperiodic 1/f background of the signal, and the broadband power that goes with it, more than the specific oscillations that carry task meaning [1, 2, 3, 4]. This matters because the EEG power spectrum separates into a periodic component, the rhythms such as sensorimotor mu and beta that index function [7], and an aperiodic 1/f component whose slope tracks excitation and inhibition balance and is itself a clinical biomarker [5]. A model that reaches high accuracy through aggregate power is trustworthy on a different and more fragile basis than one grounded in the rhythm a clinician would recognize. Here we extend this line of audits to EEGPT (NeurIPS 2024), a model absent from all of them and architecturally distinct: it augments masked reconstruction with a spatio temporal representation alignment objective meant to capture consistent, high signal to noise structure [6]. EEGPT is therefore a useful case for a question that grows more pressing as these architectures proliferate, whether a deliberate change to the pretraining objective actually changes what a model encodes from the signal, or whether the reported bias persists regardless. If the bias comes from the reconstruction objective, adding representation alignment should reduce it; if it is more general, it will remain.

Methods. We audit the frozen EEGPT encoder on PhysioNet motor imagery, contrasting imagined movement against rest across 20 subjects, a strong and unambiguous mu and beta contrast. We use two read outs on the frozen 512 dimensional embeddings, which is how the model is used downstream. First, linear decoding within subject and across subjects, against Common Spatial Patterns and band power baselines. Second, representational similarity analysis (RSA) [8], which correlates the geometry of the embedding space with the geometry of per channel band power in mu, beta, and a non motor control band, and with the task label. We then add a total power control, a partial correlation that removes overall signal strength, to test whether any band specific correspondence is genuine or a shadow of aggregate power. Total power is a proxy for the aperiodic component; a full periodic and aperiodic decomposition is in progress.

Results. A real sensorimotor rhythm is present, with mu desynchronization focal over contralateral cortex and significant in the C3 time frequency map (Figure 1). EEGPT decodes imagery versus rest about as well as Common Spatial Patterns within subject (73 versus 76 percent) and best of the compared methods across subjects (58 percent, above chance). Yet its representation is organized primarily by aggregate power. The embedding geometry correlates with total signal power at Spearman 0.31, larger than with any single band. Controlling only for task, mu, beta, and the control band correlate similarly (0.18, 0.23, 0.17), which alone would suggest no specificity. Adding the total power control separates them: mu (0.085) and beta (0.117) remain significantly positive, while the non motor control band collapses to zero. The band specific structure that is invisible before the control emerges only after it, and only for the sensorimotor bands. The added alignment objective does not rescue rhythm specificity here.

Significance. These results extend the aggregate and aperiodic power bias, reported so far for pure reconstruction models, to a hybrid architecture that adds representation alignment, which suggests the bias is more general than the reconstruction objective alone. For neurotechnology this is a concrete, trust relevant characterization: EEGPT works on motor imagery, but largely for reasons other than the sensorimotor rhythm, which predicts where it may fail when broadband artifacts shift. The finding also sharpens the open question in the concurrent literature, whether representation alignment can be steered toward oscillatory rather than aperiodic structure. Ongoing work runs a periodic and aperiodic decomposition to localize the effect, a cross subject RSA to ask whether the transferable component is the rhythm or the power, and a physiologically grounded attribution follow up to move from correlation to cause [4].



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